

STRATHALLEN, United Kingdom

WMO: 031440

Lat: 56.3264N

Lon: 3.7286W

Elev: 115

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.6	(c) 24.0	(d) 18.4	(e) 14.0	(f) 25.4	(g) 21.8	(h) 16.4	(i) 29.0	(j) 30.0	(k) 45.3	(l) 26.8	(m) 46.9	(n) 1.3	(o) 240	(p) 0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.6	(c) 73.2	(d) 63.3	(e) 70.0	(f) 61.2	(g) 67.4	(h) 59.6	(i) 64.8	(j) 70.9	(k) 62.8	(l) 67.9	(m) 61.1	(n) 65.6	(o) 7.7	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	84.4	67.4	60.6	79.3	65.2	59.0	74.9	63.7	29.9	71.0	28.4	68.1	27.2	65.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.2	(b) 20.3	(c) 18.0	(e) 15.3	(f) 78.5	(g) 5.2	(h) 3.5	(i) 11.5	(j) 81.0	(k) 8.5	(l) 83.1	(m) 5.5	(n) 85.1	(o) 1.7	(p) 87.6		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			15.3	78.5	5.2	3.5	11.5	81.0	8.5	83.1	5.5	85.1	1.7	87.6		
			15.6	68.8	6.1	3.2	11.2	71.2	7.7	73.1	4.3	74.9	-0.1	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	38.2	39.0	41.7	45.8	50.3	55.3	58.8	58.2	54.5	48.6	42.6	38.1
	DBStd	9.09	6.05	5.63	5.24	4.75	4.92	3.96	3.85	3.87	4.15	5.21	5.98	7.50
	HDD50	1837	367	308	262	141	55	5	0	1	10	88	230	370
	HDD65	6347	831	727	724	576	455	293	197	214	316	508	673	833
	CDD50	971	0	1	3	15	65	163	272	254	145	45	7	1
	CDD65	6	0	0	0	0	0	0	4	2	0	0	0	0
	CDH74	66	0	0	0	0	3	10	36	17	0	0	0	0
	CDH80	3	0	0	0	0	0	1	2	0	0	0	0	0
(14) Wind	WSAvg	7.1	7.9	8.2	8.6	7.7	6.9	6.6	6.1	6.0	6.6	6.9	7.0	7.2
(15) Precipitation	PrecAvg	39.2	4.9	3.1	2.4	2.4	2.8	2.6	3.1	3.3	2.6	4.0	3.9	4.2
	PrecMax	50.5	11.1	7.5	5.4	4.6	5.2	4.7	6.1	7.4	6.0	7.0	7.7	11.1
	PrecMin	25.2	1.3	0.4	0.5	0.7	0.9	0.9	0.7	0.9	0.5	0.4	1.3	1.1
	PrecStd	5.7	2.4	1.8	1.1	1.0	1.1	1.1	1.5	1.7	1.4	1.9	1.7	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	54.7	58.5	64.9	72.7	74.4	78.3	76.4	70.5	62.9	57.8	55.1
		MCWB	50.7	50.7	50.3	54.2	61.5	63.4	66.1	67.2	61.3	57.7	54.1	52.3
	2%	DB	51.3	52.1	54.4	59.6	66.7	69.6	73.7	71.6	66.6	60.3	55.3	52.9
		MCWB	48.5	48.8	48.7	51.3	57.3	60.5	64.0	62.5	59.2	56.1	52.1	50.4
	5%	DB	49.3	50.1	52.1	56.9	62.8	66.5	70.4	68.4	64.1	58.4	53.3	50.9
		MCWB	46.6	47.0	47.4	50.1	54.3	58.4	62.1	60.5	57.5	54.6	50.5	48.5
	10%	DB	47.2	48.0	50.1	54.5	59.6	63.8	67.1	66.0	62.0	56.5	51.3	48.6
		MCWB	44.6	44.8	46.1	48.6	52.5	56.9	60.1	59.0	56.5	53.0	48.6	46.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	51.8	52.2	55.7	62.7	65.3	68.6	68.0	63.7	58.8	55.2	53.1
		MCDB	53.3	53.8	56.8	61.7	71.2	72.1	74.9	75.0	68.3	61.2	57.1	54.9
	2%	WB	49.0	49.4	50.1	53.0	58.5	61.8	65.3	64.1	60.9	56.9	52.8	50.9
		MCDB	51.0	51.6	53.3	58.1	65.3	67.5	72.0	69.2	64.7	59.5	54.7	52.7
	5%	WB	47.0	47.5	48.2	51.0	55.7	59.7	63.2	62.1	59.1	55.3	50.9	48.7
		MCDB	49.0	49.9	51.3	55.7	61.4	65.0	68.6	66.4	62.6	57.8	52.8	50.7
	10%	WB	44.9	45.3	46.5	49.3	53.6	57.8	61.3	60.6	57.5	53.7	49.1	46.5
		MCDB	47.1	47.8	49.7	53.5	58.3	62.5	65.9	64.4	60.9	56.1	51.1	48.5
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.9	12.5	14.3	16.0	15.2	15.6	15.8	14.4	12.4	10.9	10.2
		MCDBR	10.7	11.8	15.0	18.6	21.9	20.8	21.2	19.5	17.6	13.3	11.7	11.5
		MCWBR	9.6	9.6	10.5	11.8	13.2	12.9	12.5	11.6	11.1	9.8	10.1	10.5
	5% WB	MCDBR	10.7	10.7	13.2	16.6	19.3	18.9	18.3	16.8	14.5	12.0	11.2	11.4
		MCWBR	10.0	9.3	10.1	11.2	12.5	12.9	12.0	11.1	10.2	9.7	10.3	10.7
(40) Clear-Sky Solar Irradiance	taub		0.321	0.336	0.366	0.388	0.390	0.388	0.388	0.371	0.358	0.341	0.315	0.304
	taud		2.222	2.319	2.270	2.247	2.290	2.325	2.330	2.385	2.430	2.411	2.310	2.195
	Ebn at Noon		176	228	249	262	269	271	268	266	252	224	180	149
	Edh at Noon		18	24	32	37	38	37	36	32	27	22	16	14
(44) All-Sky Solar Radiation	RadAvg		152	357	683	1125	1493	1465	1409	1123	786	422	199	113
	RadStd		23	36	78	113	105	133	129	82	54	51	20	14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page